

Ans. to the Q. No-01

(a)

here,

$$P(\text{pass} \cap \text{cheat} \mid \text{study}) = \frac{P(\text{pass} \cap \text{cheat} \cap \text{study})}{P(\text{study})}$$

$$= \frac{0.25}{0.25 + 0.15 + 0.02 + 0.03}$$

$$= \frac{5}{9}$$

again,

$$P(\text{pass} \mid \text{study}) * P(\text{cheat} \mid \text{study})$$

$$= \frac{P(\text{pass} \cap \text{study})}{P(\text{study})} * \frac{P(\text{cheat} \cap \text{study})}{P(\text{study})}$$

$$= \frac{0.25 + 0.15}{0.25 + 0.15 + 0.02 + 0.03} * \frac{0.25 + 0.02}{0.25 + 0.15 + 0.02 + 0.03}$$

$$= \frac{8}{9} * \frac{3}{5}$$

$$= \frac{24}{45}$$

$$= \boxed{\frac{8}{15}}$$

So, $P(\text{pass} \cap \text{cheat} \mid \text{study}) \neq P(\text{pass} \mid \text{study}) * P(\text{cheat} \mid \text{study})$

Therefore, cheat and pass are not conditionally independent given study.

⑥

$$\begin{aligned}
 P(\text{pass or cheat}) &= P(\text{pass} \cup \text{cheat}) \\
 &= P(\text{pass}) + P(\text{cheat}) - P(\text{pass} \cap \text{cheat}) \\
 &= (0.25 + 0.15 + 0.10 + 0.13) + (0.25 + 0.02 + 0.10 + 0.22) - (0.25 + 0.10) \\
 &= \frac{63}{100} + \frac{59}{100} - \frac{7}{20} \\
 &= \frac{87}{100} \\
 &= \boxed{0.87}
 \end{aligned}$$

Ans. to the Q. No-02

② the marginal probability of ~cold is,

$$\begin{aligned}
 P(\sim \text{Cold}) &= 0.12 + 0.04 + 0.10 + 0.07 \\
 &= \boxed{0.33}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{5} P(\sim \text{cloudy} | \sim \text{rain} \cap \sim \text{cold}) &= \frac{P(\sim \text{cloudy} \cap \sim \text{rain} \cap \sim \text{cold})}{P(\sim \text{rain} \cap \sim \text{cold})} \\
 &= \frac{0.07}{0.04 + 0.07} \\
 &= \frac{7}{11} \\
 &= \boxed{0.6364}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{c} P(\sim \text{rain} | \sim \text{cloudy}) &= \frac{P(\sim \text{rain} \wedge \sim \text{cloudy})}{P(\sim \text{cloudy})} \\
 &= \frac{0.03 + 0.07}{0.26 + 0.03 + 0.10 + 0.07} \\
 &= \frac{5}{23} \\
 &= \boxed{0.2174}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{d} P(\sim \text{rain} \vee \text{cloudy}) &= P(\sim \text{rain}) + P(\text{cloudy}) - P(\sim \text{rain} \wedge \text{cloudy}) \\
 &= (0.06 + 0.04 + 0.03 + 0.07) + (0.22 + 0.12 + 0.06 + 0.04) - (0.06 + 0.04) \\
 &= \frac{1}{5} + \frac{27}{50} - \frac{1}{10} \\
 &= \frac{16}{25} \\
 &= \boxed{0.64}
 \end{aligned}$$

Ans. to the Q. No-03

$$\textcircled{a} P(\text{playing football} | \text{left handed person}) = \frac{0.15}{0.24 + 0.15 + 0.15} \\ = \boxed{0.278}$$

$$\textcircled{b} P(\text{right handed person} | \text{playing cricket}) = \frac{0.1}{0.24 + 0.1} \\ = \boxed{0.2941}$$

$$\textcircled{c} P(\text{football} \cap \text{cricket}) = P(\text{football}) * P(\text{cricket}) \\ = (0.15 + 0.1) * (0.1 + 0.29) \\ = \boxed{0.085}$$

$$\textcircled{d} P(\text{right handed} \cup \text{left handed}) \\ = P(\text{right handed}) + P(\text{left handed}) \\ = (0.1 + 0.1 + 0.26) + (0.24 + 0.15 + 0.15) \\ = \boxed{1}$$

$$\textcircled{e} P(\text{Football} \cap \text{Right handed}) = \boxed{0.1}$$

$$P(\text{football}) * P(\text{right handed}) = (0.15 + 0.1) * (0.1 + 0.1 + 0.26) \\ = \boxed{0.115}$$

So,

$$P(\text{football} \cap \text{right handed}) \neq P(\text{football}) * P(\text{right handed})$$

$$\text{now } P(\text{playing football} \mid \text{right handed person})$$

$$= \frac{P(\text{football} \wedge \text{right handed})}{P(\text{right handed})}$$

$$= \frac{0.1}{0.1 + 0.1 + 0.26}$$

$$= \frac{5}{23}$$

$$= \boxed{0.2174}$$

So playing football depends on being right handed.

here, Ans. to the Q. No-04

$$P(\text{correctly detect} \mid \text{liar}) = 0.96$$

$$\therefore P(\neg \text{correctly detect} \mid \text{liar}) = 1 - 0.96 = 0.04$$

$$P(\text{correctly detect} \mid \neg \text{liar}) = 0.95$$

$$P(\neg \text{correctly detect} \mid \neg \text{liar}) = 1 - 0.95 = 0.05$$

$$P(\text{liar}) = 0.02$$

$$P(\neg \text{liar}) = 1 - 0.02 = 0.98$$

$$\begin{aligned} \textcircled{a} P(\text{correctly detect} \wedge \text{liar}) &= P(\text{correctly detect} \mid \text{liar}) * P(\text{liar}) \\ &= 0.96 \times 0.02 \\ &= 0.0192 \end{aligned}$$

(6)

$$\therefore P(\text{correctly detect} \wedge \neg \text{liar}) = P(\text{correctly detect} | \neg \text{liar}) \cdot P(\neg \text{liar})$$

$$= 0.95 \times 0.98$$

$$= 0.931$$

$$\therefore P(\text{correctly detect}) = P(\text{correctly detect} \wedge \text{liar}) + P(\text{correctly detect} \wedge \neg \text{liar})$$

$$= 0.0192 + 0.931$$

$$= 0.9502$$

$$\therefore P(\text{liar} | \text{correctly detect}) = \frac{P(\text{liar} \wedge \text{correctly detect})}{P(\text{correctly detect})}$$

$$= \frac{0.0192}{0.9502}$$

$$\therefore P(\neg \text{liar} | \text{correctly detect}) = \frac{P(\neg \text{liar} \wedge \text{correctly detect})}{P(\text{correctly detect})}$$

$$= \frac{0.931}{0.9502} \approx 0.98$$

So,

$$P(\text{liar} | \text{correctly detect}) < P(\neg \text{liar} | \text{correctly detect})$$

Therefore, 98% (approximately) it's more likely that the person is not a liar.

$$⑥ P(\text{correctly detect} \wedge \text{liar}) = P(\text{correctly detect} | \text{liar}) * P(\text{liar})$$

$$= 0.95 \times 0.98$$

$$= 0.931$$

$$P(\text{correctly detect} \wedge \text{liar}) = P(\text{correctly detect} | \text{liar}) * P(\text{liar})$$

$$= 0.96 \times 0.02$$

$$= 0.0192$$

$$\therefore P(\text{correctly detect}) = P(\text{correctly detect} \wedge \text{liar}) + P(\text{correctly detect} \wedge \text{liar})$$

$$= 0.0192 + 0.931$$

$$= 0.9502$$

$$\therefore P(\text{liar} | \text{correctly detect}) = \frac{P(\text{liar} \wedge \text{correctly detect})}{P(\text{correctly detect})}$$

$$= \frac{0.931}{0.9502}$$

$$= 0.98$$

Ans. to the Q. No-05

The problem of Bayes theorem is that it can be difficult to calculate the probability of an event occurring given that there another event has already occurred. Because the theorem requires the prior probability of both events where the prior probability is the probability of an event occurring without any information.

Whereas, Naive Bayes is a method of solving the problem of Bayes theorem by making the assumption that the events are independent of each other. This means that the probability of one event occurring does not affect the probability of the other event occurring. This assumption may is not true always but it often makes the calculation of probabilities easier.

For example, we have a dataset of 10 days' weather with specific characters (sunny, dry, windy) was good to play tennis or not.

Now we want to compute $P(A|B \wedge C \wedge D)$ where

"A" is play tennis, "B" is "warm" day, "C" is "dry" day and "D" is windy day.

Now, to apply Bayes' Theorem, we would need to compute $P(A)$ but also $P(B \cap C \cap D | A)$. But computing this last probability is difficult and makes it hard to apply directly Bayes' Theorem. Because computing this probability with some accuracy requires an extensive record and, in many cases, is not even possible.

However, through Naive Bayes classification, the assumption that all input features are conditionally independent like,

$$P(B \cap C \cap D | A) = P(B | A) P(C | A) P(D | A)$$

which is easy to compute. Since each of these factors can be easily estimated from the dataset.

Ans to the Q.No-06

(a)

here,

$$P(\text{play tennis} \mid \text{outlook} = \text{Sunny} \wedge \text{humidity} = \text{Mild} \wedge \text{temp} = \text{Normal} \wedge \text{windy} = \text{TRUE}) = ?$$

now,

$$P(\text{outlook} = \text{Sunny} \mid \text{play tennis}) = \frac{2}{5}$$

$$P(\text{humidity} = \text{Mild} \mid \text{play tennis}) = \frac{3}{5}$$

$$P(\text{temp} = \text{Normal} \mid \text{play tennis}) = \frac{4}{5}$$

$$P(\text{windy} = \text{TRUE} \mid \text{play tennis}) = \frac{3}{5}$$

$$P(\text{play tennis}) = \frac{5}{8}$$

$$P(\text{outlook} = \text{Sunny} \mid \neg \text{play tennis}) = \frac{3}{3} = 1$$

$$P(\text{humidity} = \text{Mild} \mid \neg \text{play tennis}) = \frac{2}{3}$$

$$P(\text{temp} = \text{Normal} \mid \neg \text{play tennis}) = \frac{0}{3} = 0$$

$$P(\text{windy} = \text{TRUE} \mid \neg \text{play tennis}) = \frac{1}{3}$$

$$P(\neg \text{play tennis}) = \frac{3}{8}$$

Decision making

$$P(\text{play tennis} \mid \text{outlook} = \text{Sunny} \wedge \text{humidity} = \text{Mild} \wedge \text{temp} = \text{Normal} \wedge \text{windy} = \text{TRUE}) = \frac{2}{5} \times \frac{3}{5} \times \frac{4}{5} \times \frac{3}{5} \times \frac{5}{8} = 0.072$$

and

$$P(\neg \text{play tennis} | \text{outlook} = \text{Sunny} \wedge \text{humidity} = \text{Mild} \wedge \text{temp} = \text{Normal} \wedge \text{windy} = \text{TRUE}) = 1 \times \frac{2}{3} \times 0 \times \frac{1}{3} \times \frac{3}{8} = 0$$

So, Player is going to play tennis given such condition because $[0.072 > 0]$..

(b)

$$P(\text{outlook} = \text{overcast} | \text{play tennis}) = \frac{2}{5}$$

$$P(\text{humidity} = \text{hot} | \text{play tennis}) = \frac{0}{5} = 0$$

So,

$$P(\text{play tennis} | \text{outlook} = \text{overcast} \wedge \text{humidity} = \text{hot} \wedge \text{temp} = \text{Normal} \wedge \text{windy} = \text{TRUE}) = \frac{2}{5} \times 0 \times \frac{4}{5} \times \frac{3}{5} \times \frac{5}{8} = 0$$

So, Player is not going to play tennis given scenario.

Ans to the Q. NO-07.

① here, considering "Edible" as class.
total No. of "yes" = 9

$$\therefore P(\text{yes}) = 9/16$$

total No. of "No" = 7

$$\therefore P(\text{no}) = 7/16$$

we know,

Entropy formula = $\sum_{i=1}^n -P_i \log P_i$ [$i \Rightarrow$ number of labels]

So,

$$\begin{aligned} E(\text{Edible}) &= -P(\text{yes}) \log_2 P(\text{yes}) - P(\text{no}) \log_2 P(\text{no}) \\ &= -9/16 \log_2 (9/16) - 7/16 \log_2 (7/16) \\ &= \boxed{0.989} \end{aligned}$$

⑥

Color	yes	no	Total
Yellow	8	5	13
Green	1	2	3

Size	yes	no	Total
Small	6	2	8
Large	3	5	8

Shape	yes	no	total
Round	6	6	12
Irregular	3	1	4

$$\begin{aligned} \therefore IG(\text{Color}) &= 0.989 - \left[\frac{13}{16} \times \left(-\frac{8}{13} \log_2\left(\frac{8}{13}\right) - \frac{5}{13} \log_2\left(\frac{5}{13}\right) \right) + \right. \\ &\quad \left. \frac{3}{16} \times \left(-\frac{1}{3} \log_2\left(\frac{1}{3}\right) - \frac{2}{3} \log_2\left(\frac{2}{3}\right) \right) \right] \\ &= 0.03 \end{aligned}$$

$$\begin{aligned} IG(\text{Size}) &= 0.989 - \left[\frac{8}{16} \times \left(-\frac{6}{8} \log_2\left(\frac{6}{8}\right) - \frac{2}{8} \log_2\left(\frac{2}{8}\right) \right) + \frac{8}{16} \times \left(-\frac{3}{8} \log_2\left(\frac{3}{8}\right) - \frac{5}{8} \log_2\left(\frac{5}{8}\right) \right) \right] \\ &= 0.11 \end{aligned}$$

$$\begin{aligned} IG(\text{Shape}) &= 0.989 - \left[\frac{12}{16} \times \left(-\frac{6}{12} \log_2\left(\frac{6}{12}\right) - \frac{6}{12} \log_2\left(\frac{6}{12}\right) \right) + \right. \\ &\quad \left. + \frac{4}{16} \times \left(-\frac{3}{4} \log_2\left(\frac{3}{4}\right) - \frac{1}{4} \log_2\left(\frac{1}{4}\right) \right) \right] \\ &= 0.04 \end{aligned}$$

So, "Size" is having highest information gain.

$\therefore$ Size is the better feature.